SATELLITE INTERNET

WIRED VIA SPACE

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India has over a billion internet subscribers, second only to China. But this number hides a stark gap: rural India has barely 46 connections per 100 people, leaving millions offline. The fix may lie above—literally. Satellite internet is being hailed as the bridge to universal connectivity. On Nov. 5, Maharashtra became the first state to sign up with Starlink—the world's largest satellite internet provider—to bring connectivity to schools, healthcare facilities and critical public infrastructure in remote areas, ahead of the company's India launch pending final approvals. Here's a snapshot of the technology that could reshape India's digital map.

1,003 mn
TOTAL INTERNET SUBSCRIBERS IN INDIA

958 mn WIRELESS
45 mn WIRED

580 mn URBAN
423 mn RURAL

70.87
INTERNET SUBSCRIBERS/ 100 POPULATION

113.83 URBAN
46.73 RURAL

HOW IT WORKS

- Traditional broadband thrives in cities, but rural and remote areas often lag; satellite internet can bridge this gap
- It relies on constellations of hundreds to thousands of **small satellites in Low Earth Orbit (LEO)** to relay data between a user antenna and ground station
- Their **low altitude** keeps latency minimal, enabling smooth video calls, streaming and real-time connectivity
- In contrast, Geostationary (GEO) satellites, fixed 36,000 km above Earth, support communications, but their distance creates noticeable lag
- Medium Earth Orbit (MEO) satellites sit in between, powering systems like GPS

